large numbers of people, these little worlds would somehow have to be fixed.

In “Collision Orbit,” Williamson portrays a group of “spatial engineers,” able to render such barren outposts clement. Coining a word, Williamson called the process of metamorphosis into an Earth-like world “terraforming.” He knew that the low gravity on an asteroid means that any atmosphere generated or transported there would quickly escape to space. So his key terraforming technology was “paragravity,” an artificial gravity that would hold a dense atmosphere.

As nearly as we can tell today, paragravity is a physical impossibility. But we can imagine domed, transparent habitats on the surfaces of asteroids, as suggested by Konstantin Tsiolkovsky, or communities established in the *insides* of asteroids, as outlined in the 1920s by the British scientist J. D. Bernal. Because asteroids are small and their gravities low, even massive subsurface construction might be comparatively easy. If a tunnel were dug clean through, you could jump in at one end and emerge some 45 minutes later at the other, oscillating up and down along the full diameter of this world indefinitely. Inside the right kind of asteroid, a carbonaceous one, you can find materials for manufacturing stone, metal, and plastic construction and plentiful water—all you might need to build a subsurface closed ecological system, an underground garden. Implementation would require a significant step beyond what we have today, but—unlike “paragravity”—nothing in such a scheme seems impossible. All the elements can be found in contemporary technology. If there were sufficient reason, a fair number of us could be living on (or in) asteroids by the twenty-second century.

They would of course need a source of power, not just to sustain themselves, but, as Bernal suggested, to move their asteroidal homes around. (It does not seem so big a step from explosive alteration of asteroid orbits to a more gentle means of propulsion a century or two later.) If an oxygen atmosphere were generated from chemically bound water, then organics could be burned to generate power, just as fossil fuels are burned on the Earth today. Solar power could be considered, although for the main-belt asteroids the intensity of